

Solution

SEC - A

Q1 a) $W_A = W_B = W_C = W_D = 40\lambda = 40 \times 90\text{ nm} = 3600\text{ nm}$.

b) $F = \frac{A \cdot B + C \cdot D}{\dots}$

c) $C_{sb} = \text{Source area} \times C_j + \text{Source perimeter} \times C_{jsw}$
 source area $= 2[5 \times 40\lambda^2] + 6 \times 40\lambda^2 = 640\lambda^2$
 source perimeter $= 2 \times 50\lambda + 6 \times 2\lambda = 112\lambda$.
 $2\lambda = 180\text{ nm} \Rightarrow \lambda = 90\text{ nm} = 0.09\mu\text{m}$.

$$C_{sb} = 640 \times (0.09)^2 \mu\text{m}^2 \times 3 \times 10^{-4} \text{ pF}/\mu\text{m}^2 + 112 \times 0.09 \mu\text{m} \times 2 \times 10^{-4} \text{ pF}/\mu\text{m}$$

$$C_{sb} = (5.184 \times 3 \times 10^{-4} + 10.08 \times 2 \times 10^{-4}) \text{ pF}$$

$$C_{sb} = 35.712 \times 10^{-4} \text{ pF}$$

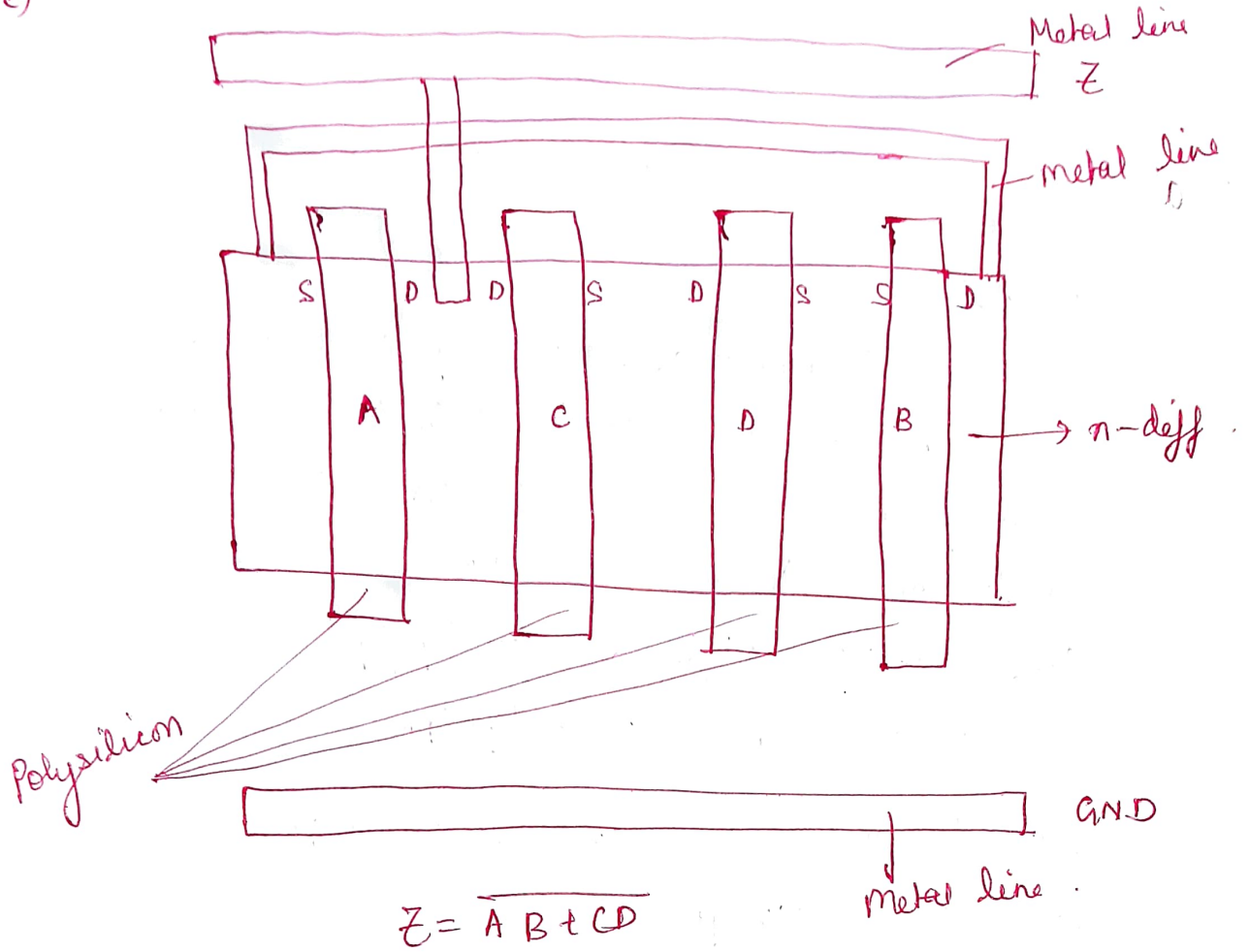
d) $C_{db} = \text{Drain area} \times C_j + \text{Drain perimeter} \times C_{jsw}$
 Drain area $= 2 \times 6 \times 40\lambda^2 = 480\lambda^2$
 Drain perimeter $= 2 \times 12\lambda = 24\lambda$

$$C_{db} = 480 \times (0.09)^2 \mu\text{m}^2 \times 3 \times 10^{-4} \text{ pF}/\mu\text{m}^2 + 24 \times 0.09 \mu\text{m} \times 2 \times 10^{-4} \text{ pF}/\mu\text{m}$$

$$C_{db} = (11.664 \times 10^{-4} + 4.32 \times 10^{-4}) \text{ pF}$$

$$C_{db} = 15.984 \times 10^{-4} \text{ pF}$$

e)



Q.2

CMOS inverter

$$K_n = 0.2 \text{ mA/V}^2, \quad K_p = 0.1 \text{ mA/V}^2$$

$$\frac{K_n}{K_p} = \frac{0.2}{0.1} = 2$$

$$|V_{tn}| = |V_{tp}| = 0.6 \text{ V}, \quad V_{DD} = 3.3 \text{ V}$$

(a) Inverter switching threshold: $V_m =$

$$V_m = \frac{V_{DD} - |V_{tp}| + V_{tn} \sqrt{\frac{K_n}{K_p}}}{1 + \sqrt{K_n/K_p}} = \frac{3.3 - 0.6 + 0.6\sqrt{2}}{1 + \sqrt{2}} = \boxed{1.97 \text{ V}}$$

(b) Resistance of each transistor in saturation

$$I_{\text{sat}} = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L}\right)_N (V_{GS} - V_{tn})^2$$

$$R_N = \frac{\partial V_{GS}}{\partial I_{\text{sat}}}; \quad V_{GS} = V_{DD}$$

differentiate

$$\frac{\partial I_{\text{sat}}}{\partial V_{GS}} = \frac{1}{R_N} = \underbrace{\mu_n C_{ox} \left(\frac{W}{L}\right)_N}_{K_n} (V_{GS} - V_{tn})$$

$$\Rightarrow R_N = \frac{1}{\left(0.2 \frac{\text{mA}}{\text{V}^2}\right) (3.3 - 0.6)} = \boxed{1.85 \text{ k}\Omega}$$

$$R_p = \frac{1}{K_p (|V_{GS}| - |V_{tp}|)} = \frac{1}{0.1 \times 2.7} = \boxed{3.7 \text{ k}\Omega}$$

(c) rise/fall time $\rightarrow$ (10% to 90%)

$$V_{\text{out}} = V_{DD} (1 - e^{-\Delta t / R_p C_L})$$

$$\Delta t \rightarrow t_r = t_1 - t_2$$

$$V_{\text{out} 10\%} = V_{DD} (1 - e^{-t_1 / R_p C_L})$$

$$V_{\text{out} 90\%} = V_{DD} (1 - e^{-t_2 / R_p C_L})$$

$$\text{for } V_{\text{out}} = 10\% \text{ to } 90\%$$

$$\Delta t = t_r = 2.2 R_p C_L$$

$$t_r = (2.2) (3.7 \text{ k}) \times 9 \text{ fF} = \boxed{73.3 \text{ pS}}$$

$$t_f = 2.2 R_N C_L = 2.2 \times 1.85 \text{ k} \times 9 \text{ fF} = \boxed{36.7 \text{ pS}}$$

(d)

(d) Propagation Delays $\rightarrow$

$$\tau_{PLH} = 0.69 (R_D) (9+25) fF$$
$$= 0.69 \times 3.7K \times 34 fF$$

$$\tau_{PLH} = \boxed{87.3 pS}$$

$$\tau_{PHL} = 0.69 \times \underbrace{1.85K}_{R_n} \times 34 fF = \boxed{43.6 pS}$$

$$\tau_{av} = \frac{87.3 + 43.6 pS}{2} = 65.45 pS$$

(e) Ring Oscillator (11 stages)

$$f_{req} \quad f = \frac{1}{2\tau_p \times N} = \frac{1}{2 \times 65.45 p \times 11} = \boxed{694 MHz}$$

Power delay product
of one stage = $C_L V_{DD}^2 f$

$$(34 fF) \times (3.3)^2 \times 694 MHz$$

(dyn. power consumption) $= \boxed{256.96 \mu W}$

SOLUTIONS

Question 3:

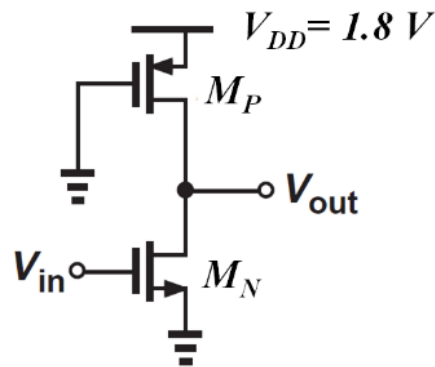


Figure 1(a)

a). Topology shown is $\Rightarrow$ Pseudo N-MOS Inverter Circuit.

M_P is operating in saturation region.

M_N is operating in linear region ($V_{OL} = 0.1V$)

$$\therefore V_{in} = V_{DD}$$

$$I_{DP} = \frac{1}{2} \mu_p C_{ox} \left(\frac{W}{L} \right)_{M_P} (V_{GS} - V_{T_{DP}})^2$$

$$= \frac{1}{2} \times 50 \times 10^{-6} \times \left(\frac{3}{0.18} \right) \times (1.8 - 0.5)^2$$

$$\boxed{I_{DP} = 7.04 \times 10^{-4} A}$$

$$\text{Also, } I_{DP} = I_{DN} = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right)_{M_N} [2(V_{GS} - V_{T_{DN}}) V_{DS} - V_{DS}^2]$$

$$\Rightarrow 7.04 \times 10^{-4} = \frac{1}{2} \times 100 \times \left(\frac{W}{L} \right)_{M_N} [2(1.8 - 0.4) \times 0.1 - (0.1)^2]$$

$$\Rightarrow \boxed{\left(\frac{W}{L} \right)_{M_N} = 52.16}$$

b). Without parasitic resistance $\Rightarrow$ Normal CMOS Inverter

(i). $V_{OL} = 0V$ & $V_{OH} = V_{DD} = 1.8V$

$$(ii). V_{IL} = \frac{2V_{out} + V_{TOP} - V_{DD} + K_R V_{TON}}{1 + K_R}$$

$$K_R = \frac{\mu_n C_{ox} \left(\frac{W}{L}\right)_{MN}}{\mu_p C_{ox} \left(\frac{W}{L}\right)_{MP}} = \frac{100 \times 10^{-6} \times \frac{0.5}{0.18}}{50 \times 10^{-6} \times \frac{11}{0.18}} = \frac{22}{11} = \frac{10}{11}$$

$$V_{out} = V_{DD} = 1.8V \text{ (given)}$$

$$\Rightarrow V_{IL} = (2 \times 1.8 + 0.5 - 1.8 + \frac{10}{11} \times 0.4) / (1.909)$$

$$\Rightarrow \boxed{V_{IL} = 0.87V}$$

$$V_{IH} = \frac{V_{DD} + V_{TOP} + K_R (2V_{out} + V_{TON})}{1 + K_R}$$

$$= \frac{1.8 + (-0.5) + \frac{10}{11} (2 \times 0 + 0.4)}{\left(1 + \frac{10}{11}\right)} \quad \{V_{out} = 0V \text{ (given)}\}$$

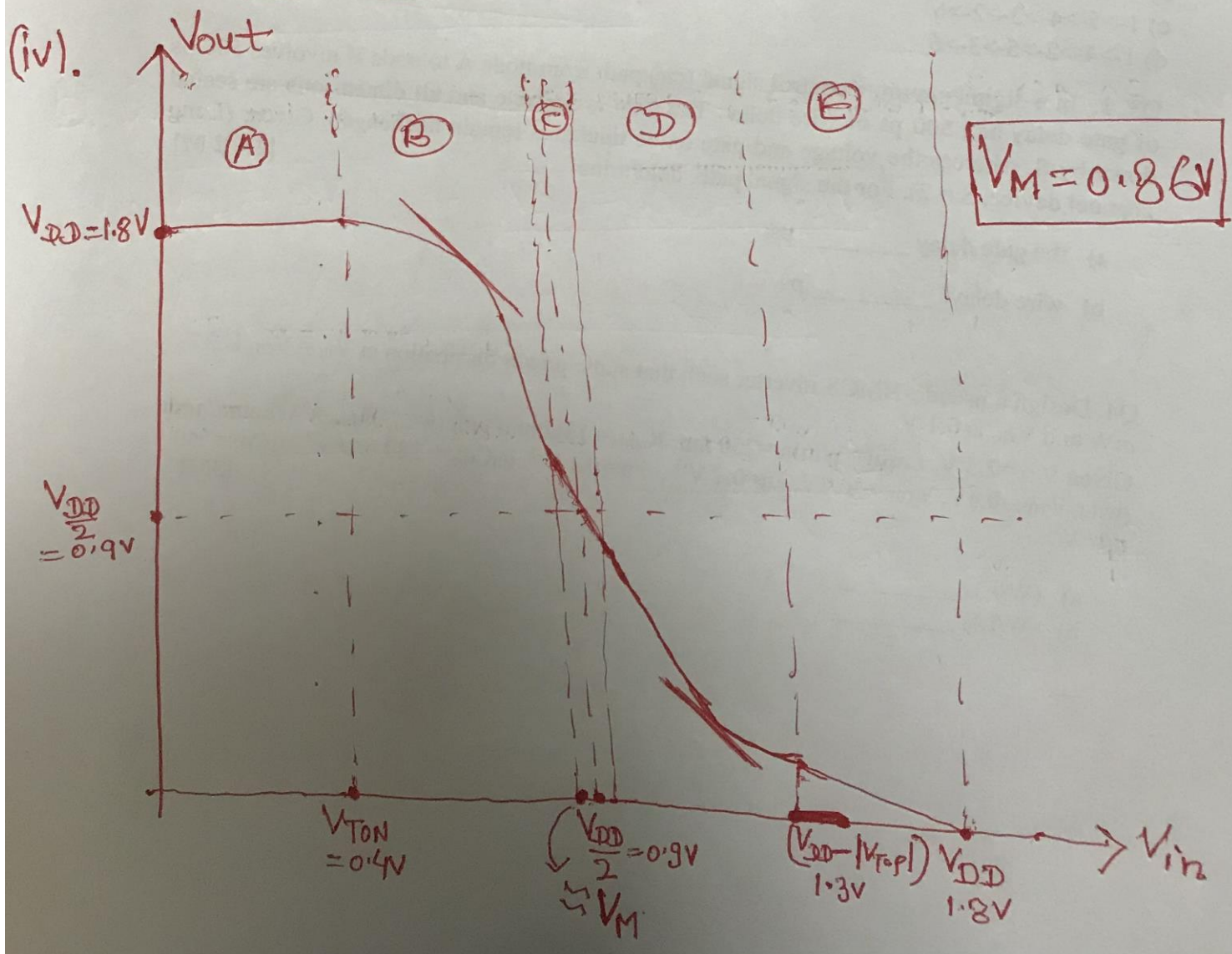
$$\Rightarrow \boxed{V_{IH} = 0.87V}$$

(iii). $NM_L = V_{IL} - V_{OL}$

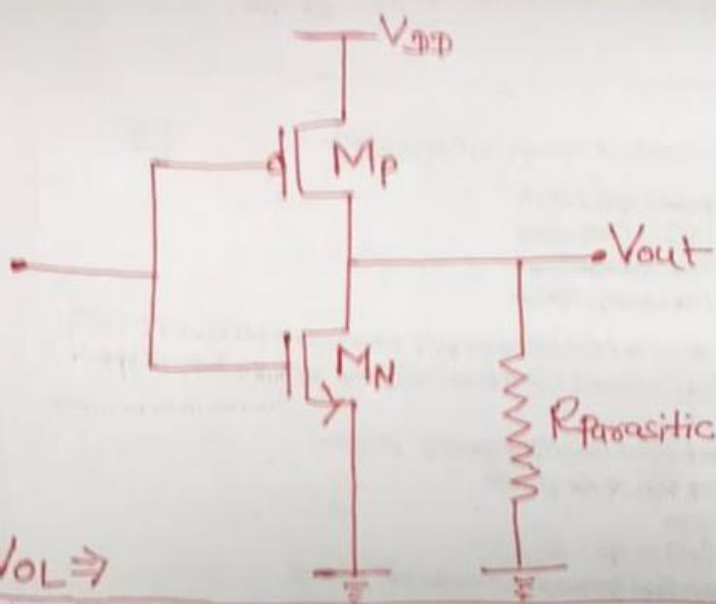
$$\Rightarrow NM_L = 0.87 - 0 = 0.87V$$

$$NM_H = V_{OH} - V_{IH}$$

$$\Rightarrow NM_H = 1.8 - 0.87 = 0.93V$$



(V). With the parasitic resistance, $R_{\text{parasitic}}$:



$V_{OL} \Rightarrow$

$V_{OL} = 0V$, since MOS transistor M_p is off for $V_{in} = V_{DD}$

$V_{OH} \Rightarrow$

$V_{OH} = V_{DD} - \Delta V$, because even if M_N is off for $V_{in} = 0V$ (say), there will be current flow through $R_{\text{parasitic}}$.

Hence, output node will not get charged upto V_{DD} .

When, M_N is off for $V_{in} = 0V$, $I_{D_{Rp}} = \frac{V_{out}}{R_{\text{parasitic}}}$

$$\Rightarrow \frac{V_{out}}{R_{\text{parasitic}}} = \frac{1}{2} \mu_p C_{ox} \left(\frac{W}{L} \right)_{M_p} \left[2(V_{DD} - V_{Thp}) (V_{DD} - V_{out}) - (V_{DD} - V_{out})^2 \right]$$

$$\Rightarrow V_{out} = V_{OH} = 1.348V$$

SECTION-B

4. a) $G_1 = (A_0 \odot B_0) \cdot (A_1 \odot B_1)$

b) No, static gates must be in even numbers between two domino gates.

c)

I. $A_0=1, B_0=0, A_1=1, B_1=0$

$$\text{node voltage at P } (V_P) = \frac{V_{DD}}{1 + \frac{C_x}{C_P}} = \frac{2.5}{35} \times 30$$
$$= 2.14V > \text{higher than } V_m$$

$$V_F=0 \Rightarrow V_D = 2.5V \Rightarrow V_G = 0V$$

$$G_1 = (A_0 \odot B_0) \cdot (A_1 \odot B_1) = 0 \rightarrow \text{As per the logic}$$

II. $A_0=1, B_0=1, A_1=1, B_1=0$

$$V_P=0 \Rightarrow V_F = 2.5V$$

$$\text{Now } V_D = \frac{2.5}{24+15+10} \times 24 = 1.22V < V_m(1.5V)$$

hence, $V_{G_1} = 2.5V$

$$G_1 = (A_0 \odot B_0) \cdot (A_1 \odot B_1) = 1 \cdot 0 = 0$$

(d) Logic dictates node voltage to be zero
↓
Error

Remedy: Adding a weak PMOS Pullup device to the dynamic CMOS o/p.

$$e) \quad \overline{D} = NF^{\frac{1}{N}} + P$$

$$N = 4$$

$$G = \frac{3}{3} \times 1 \times \frac{4}{3} \times 1 = \frac{4}{3}$$

$$P = \frac{8 + 3 + 6 + 3}{3} = \frac{20}{3} = 6.67 \text{ d.u.}$$

$$B = 1$$

$$H = 50$$

$$F = GBH = \frac{4}{3} \times 1 \times 50 = \frac{200}{3}$$

$$\overline{D} = 4 \times \left(\frac{200}{3} \right)^{\frac{1}{4}} + 6.67$$

$$= 11.43 + 6.67 = 18.1 \text{ d.u.}$$

Soln: (i) $\text{Max } V_{id(\text{saturation})} = 0.282 \text{ V}$

$$V_{OV(m_1, m_2, m_3)} = \frac{0.282 \text{ V}}{\sqrt{2}} = 0.2 \text{ V} \quad (1M)$$

$$V_{icm(\text{min})} = V_{OV(m_1)} + V_{OV(m_2)} = 0.2 + 0.6 = 0.8 \text{ V} \quad (1M)$$

$$V_{icm(\text{max})} = 0.8 + 1 \text{ cmR} = 2.5 \text{ V} \quad (1M)$$

$$V_{icm(\text{max})} = V_{DD} - |V_{OV6}| - |V_{OV4}| + V_{th}$$

$$|V_{OV4}| = |V_{OV6}| = \frac{2.5 - 0.4 - 2.5}{2}$$

$$= 0.2 \text{ V} \quad (1M)$$

$$V_{out(OL)} = V_{DD} - V_{SG7}$$

$$= 2.5 - (0.4 + 0.2) = 1.9 \text{ V} \quad (1M)$$

$$V_{out2(\text{max})} = V_{DD} - 2|V_{OV,P}| = 2.5 - 0.4 = 2.1 \text{ V} \quad (1M)$$

$$V_{out2(\text{min})} = 2|V_{OV,n}| = 0.4 \text{ V} \quad (1M)$$

(iii) $S.R. = \frac{I_{SS}}{C_L} = \frac{200 \text{ V}}{\mu\text{s}}$

$$C_L = 0.5 \text{ pF (given)}$$

$$\Rightarrow I_{SS} = 100 \text{ nA} \quad (1M)$$

$$g_{m2} = \frac{2 \times 50 \mu\text{A}}{0.2} = 500 \mu\text{A/V} \quad (1M)$$

$$r_{O2} = r_{O4} = r_{O6} = \frac{1}{\lambda I_D} = \frac{1}{0.1 \times 50 \mu\text{A}} = 200 \text{ K} \quad (1M)$$

$$\frac{V_{out2}}{V_{in}} = 2g_{m2}(r_{O2} || g_{m4}r_{O4}r_{O6}) \approx 2g_{m2}r_{O2} = 200 \text{ V/V}$$

$$g_{m3} = \frac{2 \times 50 \mu\text{A}}{0.2} = 500 \mu\text{A/V}$$

$$\frac{V_{out3}}{V_{out2}} = \frac{r_{O3}}{1/g_{m8} + r_{O3}} = \frac{200 \text{ K}}{2 \text{ K} + 200 \text{ K}} \approx 1$$

$$\therefore \frac{V_{out3}}{V_{in}} \approx 200 \text{ V/V} \quad (2M)$$

(iv) $\omega = \frac{1}{(1/g_{m8})C_3} = \frac{g_{m8}}{C_3}$

$$\Rightarrow C_3 = \frac{500 \mu\text{A/V}}{\omega} = \frac{500}{12.5 \times 10^6} = 40 \text{ pF} \quad (3M)$$

Q6

$$V_{in} = V_{id}$$

$$V_{ov} = 0.2V, C_L = 10pF, I_{SS} = 100\mu A$$

(9) folded Cascode amplifier topology.

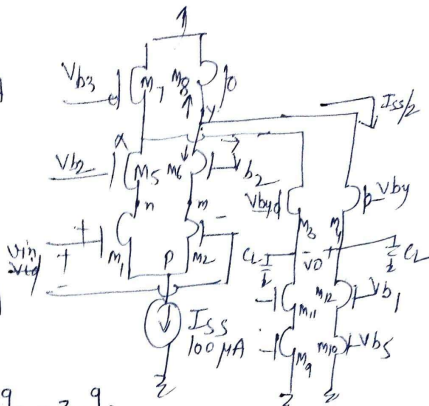
(b) Small signal DC gain $\frac{V_{out}}{V_{in}}$

$$\left(\frac{V_{out}}{V_{in}}\right) = +G_m R_{out} = 0.15 \times 10^{-3} \times \frac{2 \times 10^9}{3 \times 10^8} = 10.37dB$$

$$G_m = g_{m1,2} = 500 \mu A/V = 0.5 mA/V$$

$$R_{out} = g_{m1} r_{o1}^2 \parallel g_{m3} r_{o3} R_{yx} = (2 \times 10^9) \parallel 10^9 \Omega = \frac{2}{3} \times 10^9 \Omega$$

$$R_{yx} = r_{o8} \parallel g_{m6} r_{o6}^2 \approx 1M\Omega$$



$$\begin{aligned} g_{m1} &= g_{m2} = g_{m3} = g_{m4} \\ &= g_{m5} = g_{m6} = g_{m7,8} \\ &= g_{m1,2} = \frac{2 I_{SS/2}}{V_{ov}} = \frac{100 \mu A}{0.2} \\ &= 500 \mu A/V \end{aligned}$$

$$\begin{aligned} r_{o1} &= r_{o2} = r_{o3} = r_{o4} = r_{o5} = r_{o6} \\ &= r_{o7} = r_{o8} = r_{o9} = r_{o10} \\ r_{o1} &= \frac{1}{\lambda (I_{SS/2})} = 2M\Omega \\ r_{o7} &= r_{o8} = \frac{1}{\lambda \times I_{SS}} = 1M\Omega \end{aligned}$$

$$\begin{aligned} g_{m1} r_{o1} &= 500 \mu A/V \times 2 \times 10^6 \\ &= 1 mA/V \\ g_{m1} r_{o1} r_{o5} &= 2 \times 10^9 \Omega \\ 1/g_{m3,4} &= 2 K\Omega \end{aligned}$$

(c)

c)

Pole frequencies; — $C_1 = 10\text{pF}$

$$\omega_{xy} \approx \frac{1}{\underbrace{(R_{xy} \parallel r_{oq} \parallel \frac{1}{g_{m4}})}_{(1\text{M} \parallel 2\text{M} \parallel 2\text{k})} \underbrace{(0.5C_L)}_{(\frac{1}{2} \times 10^{-11})}} \approx \frac{10^{11}}{2\text{k} \times \frac{1}{2}} \approx 10^8 \text{ rad/sec.}$$

$$= 100 \text{ M rad/sec.}$$

$$(*) \quad |W_{out}| \approx \frac{1}{R_{out}(C_L + C_1)} \approx \frac{1}{(\frac{2}{3} \times 10^9)(15\text{pF})} = \frac{2 \times 10^6}{2 \times 10^5} = 10^5 = 100 \text{ K rad/sec}$$

(d) $U_{GB} \downarrow$

at $U_{GB} \rightarrow |A| = 0$

$$0 = 20\log|A| - 20\log \sqrt{1 + \left(\frac{U_{GB}}{\omega_x}\right)^2} - 20\log \sqrt{1 + \left(\frac{U_{GB}}{\omega_{out}}\right)^2}$$

$$= 20\log(3.3 \times 10^5) - 20\log \sqrt{1 + \left(\frac{U_{GB}}{100\text{M}}\right)^2} - 20\log \sqrt{1 + \left(\frac{U_{GB}}{100\text{k}}\right)^2}$$

$$\log 3.3 \times 10^5 \approx \log \left(\frac{U_{GB} \times U_{GB}}{100\text{M} \times 100\text{k}} \right)$$

$$(3.3 \times 10^5) = \frac{U_{GB}^2}{10 \times 10^{12}}$$

$$\Rightarrow U_{GB} =$$

$$\sqrt{3.3 \times 10^5 \times 10 \times 10^{12}}$$

$$U_{GB} = \sqrt{3.3} \times 10^9 = 1.81 \times 10^9 \text{ rad/sec.}$$

$$\boxed{U_{GB} = 1.81 \times 10^9 \text{ rad/sec}}$$

(e) Bode Plot

